

FLIPPING COINS

Suppose you have a coin with two sides

You flip the coin multiple times and get the sequence

HHT ... HTHT

=> You can formally write it as

$F_1, F_2, \dots, F_k$ where $F_i \in \{H, T\}$ $i=1 \dots k$ ($k = \#$ of coin flips)

- WHAT IS THE EXPECTED NUMBER OF COIN FLIPS UNTIL THE FIRST H APPEARS?
- ON AVERAGE over all possible sequences, WHEN DOES THE FIRST H OCCUR?

=> You have to flip the coin **2** times!

Why?

Let X be the number of coin flips until we see H

x	1	2	3	4	5	k
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{2^k}$

T first H second, both with prob 1/2

It's either H or T

Probability of x coin flips to see H

How do we compute the expected number?

$$E[X] = \sum_{i=1}^{\infty} i \cdot \frac{1}{2^i}$$

this is the number of flips times the probability that that one is the right one

How do we compute this?

$$\sum_{i=1}^{\infty} i \cdot \frac{1}{2^i} = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot \frac{1}{8} + 4 \cdot \frac{1}{16} + \dots$$

$$= 1 \cdot \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4} \right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} \right) + \left(\frac{1}{16} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16} \right) + \dots$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$$

$$\longrightarrow \sum_{i=1}^{\infty} \frac{1}{2^i}$$

$$+ \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$$

$$\longrightarrow \frac{1}{2} \sum_{i=1}^{\infty} \frac{1}{2^i}$$

$$+ \frac{1}{8} + \frac{1}{16}$$

$$\longrightarrow \frac{1}{2^2} \sum_{i=1}^{\infty} \frac{1}{2^i}$$

$$+ \frac{1}{16} \dots$$

$$\longrightarrow \frac{1}{2^3} \sum_{i=1}^{\infty} \frac{1}{2^i}$$

$$\left. \begin{array}{l} \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\sum_{i=1}^{\infty} \frac{1}{2^i} \right) \end{array} \right\}$$

$$= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{j=1}^{\infty} \frac{1}{2^j} \cdot \left(\sum_{i=1}^{\infty} \frac{1}{2^i} \right)$$

GEOMETRIC SERIES: $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q} \quad q \in (0,1)$

$$\Rightarrow \sum_{i=1}^{\infty} \frac{1}{2^i} = \sum_{i=0}^{\infty} \frac{1}{2^i} - 1 = \frac{1}{1-1/2} - 1 = 2 - 1 = 1$$

$$\begin{aligned} E[X] &= \sum_{i=1}^{\infty} \frac{1}{2^i} + \sum_{j=1}^{\infty} \frac{1}{2^j} \left(\sum_{i=1}^{\infty} \frac{1}{2^i} \right) = \\ &= 1 + \sum_{j=1}^{\infty} \frac{1}{2^j} \cdot 1 = 1 + 1 = 2 \end{aligned}$$

let X be the number of coin flips until we see HT

x	1	2	3	4	5	...
$IP(X=x)$	0	$\frac{1}{4}$	$\frac{2}{8}$	$\frac{3}{16}$	$\frac{4}{32}$	

$$E[X] = \sum_x x \cdot IP(X=x) = \sum_{i=1}^{\infty} \frac{i}{2^{i+1}}$$

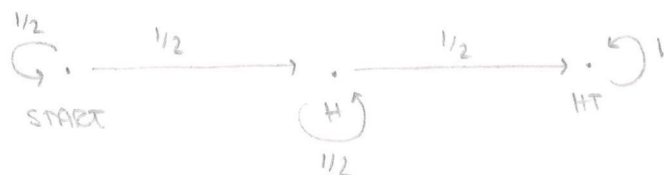
We need to flip at least two times to have two "results"

HH
HT
TT
TH
HHT
THT
HTH
TTH
HHH
TTH
HTT
TTT

lots of computations!

these two cannot be considered because we haven't have stopped already

LET'S TRY ANOTHER WAY



(MARKOV CHAIN)

Let E_x be the expected number of flips to go from state x to state HT with $x \in \{S, H, HT\}$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H$$

$\downarrow$ one flip we have to do anyway
 $\swarrow$ we flip T (we stay "not")
 $\searrow$ we move "forward" (flip H)

$$E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT}$$

$E_{HT} = 0$ (we are already where we want to "go"
i.e., no more flips needed to reach
the desired reference)

$\Rightarrow$ we have a system of linear equations!

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_H + \frac{1}{2}E_{HT} \\ E_{HT} = 0 \end{cases}$$

$$E_H = 1 + \frac{1}{2}E_H$$

$$E_H = 2$$

$$E_S = 2 + \frac{1}{2}E_S$$

$$E_S = 4 \quad \checkmark$$

we need 4 coin flips to
see HT (on average)

WHAT IS THE EXPECTED NUMBER OF FLIPS UNTIL HTH APPEARS FOR THE FIRST TIME?



ALL PROBABILITIES = $\frac{1}{2}$ (BUT THE LAST ONE)

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HT} \\ E_{HT} = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HTH} \\ E_{HTH} = 0 \end{cases}$$

$$E_{HT} = 1 + \frac{1}{2}E_S$$

$$E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(1 + \frac{1}{2}E_S\right)$$

$$= \frac{3}{2} + \frac{1}{4}E_S$$

$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(\frac{3}{2} + \frac{1}{4}E_S\right)$$

$$= 1 + \frac{3}{4} + \frac{1}{2}E_S + \frac{1}{8}E_S$$

$$1 - \frac{1}{2} - \frac{1}{8} = \frac{8-4-1}{8}$$

$$\frac{1}{8}E_S = \frac{1}{4}$$

$$E_S = 10$$

FLIPS HH

$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_S + \frac{1}{2}E_{HH} \\ E_{HH} = 0 \end{cases}$$

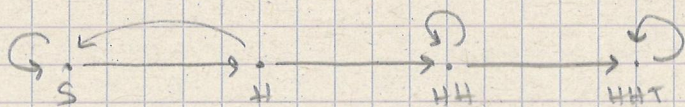
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(1 + \frac{1}{2}E_S\right)$$

$$= 1 + \frac{1}{2}E_S + \frac{1}{2} + \frac{1}{4}E_S$$

$$\left(1 - \frac{1}{2} - \frac{1}{4}\right)E_S = \frac{3}{2}$$

$$E_S = \frac{12}{2} = 6$$

FLIPS HHT



$$\begin{cases} E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}E_H \\ E_H = 1 + \frac{1}{2}E_{HH} + \frac{1}{2}E_S \\ E_{HH} = 1 + \frac{1}{2}E_{HH} + \frac{1}{2}E_{HHT} \\ E_{HHT} = 0 \end{cases}$$

$$E_{HH} = 2$$

$$E_H = 2 + \frac{1}{2}E_S$$

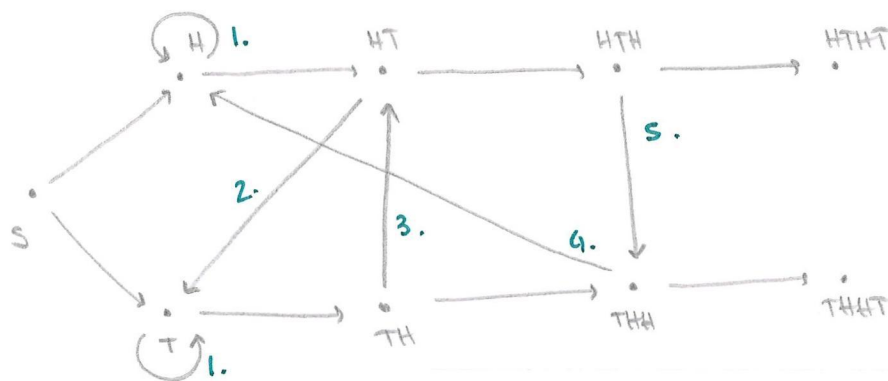
$$E_S = 1 + \frac{1}{2}E_S + \frac{1}{2}\left(2 + \frac{1}{2}E_S\right)$$

$$\left(1 - \frac{1}{2} - \frac{1}{4}\right)E_S = 2$$

$$\frac{1}{4}E_S = 2 \quad E_S = 8$$

PROBABILITY of REACHING HTHT before THHT

P_x = prob of reaching HTHT before THHT



1. If in T we flip T again is the same as if we did not flip at all. The same if we flip H in H
2. If we flip H in HT, we go "further"
If we flip T then we will have HTT which is the same as just T
3. If we flip H in TH we can continue
If we flip T we get THT and we can use HT to restart
4. If we flip T we proceed
If we flip H we get THHH and it is like we could start from H
5. If we flip T we continue
If we flip H we get HTHT and we can use THH

$$P_S = \frac{1}{2} P_H + \frac{1}{2} P_T$$

$$P_T = \frac{1}{2} P_T + \frac{1}{2} P_{TH}$$

$$P_H = \frac{1}{2} P_H + \frac{1}{2} P_{HT}$$

$$P_{TH} = \frac{1}{2} P_{HT} + \frac{1}{2} P_{THH}$$

$$P_{HT} = \frac{1}{2} P_T + \frac{1}{2} P_{HTH}$$

$$P_{HTH} = \frac{1}{2} P_{HTHT} + \frac{1}{2} P_{THH}$$

$$P_{THH} = \frac{1}{2} P_{THHT} + \frac{1}{2} P_H$$

$$P_{THHT} = 0$$

$$P_{HTHT} = 1$$

$$P_{THHT} = 0$$

$$P_{HTHT} = 1$$

$$P_{THH} = \frac{1}{2} P_H$$

$$P_{HTH} = \frac{1}{2} + \frac{1}{2} P_{THH} = \frac{1}{2} + \frac{1}{2} \left(\frac{1}{2} P_H \right) = \frac{1}{2} + \frac{1}{4} P_H$$

$$P_{HT} = \frac{1}{2} P_T + \frac{1}{2} P_{HTH} = \frac{1}{2} P_T + \frac{1}{2} \left(\frac{1}{2} + \frac{1}{4} P_H \right) = \frac{1}{2} P_T + \frac{1}{4} + \frac{1}{8} P_H$$

$$P_{TH} = \frac{1}{2} P_{HT} + \frac{1}{2} P_{THH} = \frac{1}{2} \left(\frac{1}{2} P_T + \frac{1}{4} + \frac{1}{8} P_H \right) + \frac{1}{2} \left(\frac{1}{2} P_H \right)$$

$$= \frac{1}{4} P_T + \frac{1}{8} + \frac{5}{16} P_H$$

$$P_H = \frac{1}{2} P_H + \frac{1}{2} P_{HT} = \frac{1}{2} P_H + \frac{1}{2} \left(\frac{1}{2} P_T + \frac{1}{4} + \frac{1}{8} P_H \right)$$

$$= \frac{1}{2} P_H + \frac{1}{4} P_T + \frac{1}{8} + \frac{1}{16} P_H$$

$$\frac{7}{16} P_H = \frac{1}{4} P_T + \frac{1}{8}$$

$$P_H = \frac{16}{7} \left(\frac{1}{4} P_T + \frac{1}{8} \right) = \frac{4}{7} P_T + \frac{2}{7}$$

$$P_T = \frac{1}{2} P_T + \frac{1}{2} P_{TH} = \frac{1}{2} P_T + \frac{1}{2} \left(\frac{1}{4} P_T + \frac{1}{8} + \frac{5}{16} P_H \right)$$

$$= \frac{1}{2} P_T + \frac{1}{8} P_T + \frac{1}{16} + \frac{5}{32} P_H$$

$$= \frac{5}{8} P_T + \frac{1}{16} + \frac{5}{32} \left(\frac{4}{7} P_T + \frac{2}{7} \right)$$

$$= \frac{5}{8} P_T + \frac{1}{16} + \frac{5}{56} P_T + \frac{5}{112}$$

$$\frac{2}{7} P_T = \frac{12}{112}$$

$$P_T = \frac{3}{8}$$

$$P_H = \frac{4}{7} \cdot \frac{3}{8} + \frac{2}{7} = \frac{1}{2}$$

$$P_S = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{3}{8} = \frac{7}{16}$$

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MORE NOTES ON THE AVERAGE NUMBER OF FLIPS

HHTHH

(IF WE ARE LUCKY) WE CAN GET THERE IN 5 FLIPS BUT THIS HAPPENS WITH PROBABILITY $\frac{1}{32}$ ($= \frac{1}{2^5}$).

SUPPOSE NOW WE HAVE AN INFINITE SEQUENCE OF FLIPS. WE EXAMINE THE LAST 5 FLIPS AT ANY POINT IN TIME.

$\Rightarrow$ THE CODE WE SEE IS HHTHH WITH PROBABILITY $\frac{1}{32}$

$\Rightarrow$ THE SEQUENCE APPEARS AT LEAST EVERY 32 FLIPS (I.E. APPEARS THE FIRST TIME AFTER 32 FLIPS AND REAPPEARS AFTER 32 FLIPS)

HOWEVER, IF WE START AT HHTHH WE COULD GET TO HHTHH AGAIN IN ONLY 4 FLIPS

HHTHHTHH

$\Rightarrow$ THE 32 FLIPS TO RETURN INCLUDE THE ADVANTAGE OF HAVING THE FIRST H ALREADY

BUT AT THE BEGINNING WE DO NOT HAVE H

$\Rightarrow$ WE NEED TO HAVE H FIRST (2 FLIPS)

S $\rightarrow$ H $\rightarrow$ HHTHH $\Rightarrow$ 34
2 32

HHTHH

THE SEQUENCE APPEARS EVERY 32 FLIPS (AT LEAST) BECAUSE, AS ABOVE, WE CAN OBTAIN IT WITH 5 FLIPS AND THIS HAPPENS WITH PROB $\frac{1}{32}$ ($= \frac{1}{2^5}$)

$\Rightarrow$ THESE 32 FLIPS TAKE ADVANTAGE OF THE FACT THAT WE HAVE HH ALREADY

$\Rightarrow$ BUT HH REAPPEARS IN 4 FLIPS (2^2) TAKING ADVANTAGE OF THE FACT THAT WE HAVE H ALREADY

$\Rightarrow$ BUT H REAPPEARS IN 2 FLIPS

S $\rightarrow$ H $\rightarrow$ HH $\rightarrow$ HHTHH $\Rightarrow$ 38
2 4 32

HTTHT

THE SEQUENCE APPEARS EVERY 32 FLIPS (AT LEAST) TAKING ADVANTAGE OF THE FACT THAT WE HAVE HT ALREADY

$\Rightarrow$ HT REAPPEARS EVERY 4 FLIPS

S $\rightarrow$ HT $\rightarrow$ HTTHT $\Rightarrow$ 36
4 32